

National Institute of Technology Karnataka Surathkal
Department of Information Technology



IT 252 DATABASE SYSTEMS

Introduction

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Prerequisites:

Computer Fundamentals.

Data concepts.

Data Structures & Programming.

Flow Diagrams.

Course Plan

Weeks	Topics
1	Introduction to the course, highlighting the data and databases, and related basic concepts. Advantages and need for a database system.
2	Attributes, tuples, relational schema, conceptual model, introduction to SQL-DML, DDL, creating relations/ tables, access and manipulate data.
3-4	Data model, relational model, and SQL, relational data model and relational database constraints, SQL data definition and data types, specifying constraints, basic retrieval queries, complex queries, triggers, views, schema modification.
5-6	ER model development, entity types/sets, attribute, relationship types/sets, simple employee database conceptual design using ER concepts, ER to relational mapping algorithm, EER model.
7-8	Functional dependency definition, need of normalization, anomalies and redundant information in tuples, normalization steps (1NF, 2NF, 3NF), BCNF, introduction to the higher normal forms.
9	Procedural query languages: relational algebra and SQL-SELECT, PROJECT, RENAME, binary and JOIN operations with examples. Non-Procedural query languages: relational calculus, and SQL-tuple relational

Course Plan (Contd...)

	calculus, existential and universal quantifiers with examples.
10-11	Transaction management, schedule and serializability, concurrency control, 2-phase lock, recovery mechanism: undo/redo values.
12-13	Disk storage- basic file structures, ordered/unordered. binary search, hashing, indexing, importance of indexes, primary and secondary indexing methods, clustered, B tree, B++ trees.
14	Current trends in database system, introduction to data warehousing, data mining.

Text Books and References:

1. Elmasri and S.B Navathe, Fundamentals of Database Systems,7th Ed., Pearson, 2017
2. Raghu Ramakrishnan, Johannes Gehrke, Database Management Systems, McGraw-Hill, 2014 R.
3. Silberschatz, Korth A.F., Sudarshan S., Database System Concepts, 6th Ed., McGraw-Hill,2010.

Course Assessment

Theory: 60%

20% Midsem theory exam

40% Endsem theory exam

Lab 40%

15% Regular lab

25% Mini Project- 10% Midsem Evaluation- *Proposal and Database Design*

15% Endsem Evaluation- **Database Implementation and**

Testing.

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- 1: Basic Concepts: Data, Information.
- 2: Traditional Database systems, Database Management systems (DBMS).
 - An example.
- 3: Characteristics of Database systems.
- 4: Actors on the Scene.
- 5: Workers behind the Scene.
- 6: Need and Advantages of Database approach.
- 7: When Not to Use a DBMS.

1: Introduction

Course Plan: Theory:

Basic Concepts

Week 1: *Introduction :*

Basic Concepts, Data and Databases

Need for a database system

Advantages of a database approach

Basic Concepts: Data, Information

Data:

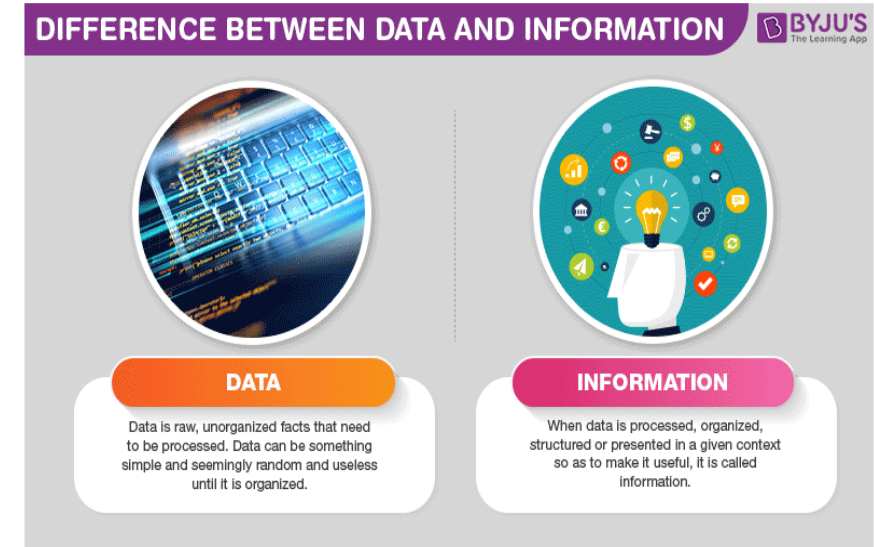
- are the **raw (unorganized) facts** about things like people, places, events or concepts that have inherent meaning, which is useless by themselves.
- Unprocessed information.



Basic Concepts: Data, Information

Information

- Processed data is known as **information**, that provides **meaning** in a specific context.
- A good decision-making needs accurate, relevant and timely **information**.
- Databases and database systems are an essential component of life in modern society: including **business**, **engineering**, **social media**, **genetics**, **banking**, **medicine**, **law**, **education**, and **library science**.



Different methods for managing and organizing data

- 1. Manual File Handling Approach**
- 2. Traditional File Based Approach**
- 3. Database Approach**

1. Manual File Handling Approach

- Physical storage, such as paper records, ledgers, or files.
- Data entry is often performed manually.
- High risk of errors due to manual processing.
- Time-consuming and inefficient.
- Difficult to scale and prone to data inconsistencies.
- Example: A small business firm manually recording transactions in a paper ledger or spreadsheet.

Limitations:

- Prone to error.
- Difficult to update, retrieve, integrate.
- You have the data but it is difficult to compile the information.
- Limited to small size information.
- Cross referencing is difficult.
- ❖ An alternative approach of data handling is a computerized way of dealing with the information. Two computerized approaches evolved:
 - ❖ File based approach → decentralized
 - ❖ Database approach → centralized

2. File Based Approach

- File based systems were an early attempt to computerize the manual filing system by use of computer system.
- This approach is the decentralized computerized data handling method.
- Such systems have number of programs for each of the different applications in the organization.
- Since every application defines and manages its own data, the system is subjected to serious data duplication.
- Example: Storing customer information in separate text or CSV files for each department (sales, marketing, etc.).

Limitations:

- Data Redundancy (Duplication of data)—Same data is held by different programs and thus wasted space.
- Separation or Isolation of Data: Available information in one application may not be known.
- Limited data sharing
- Lengthy development and maintenance time
- Data dependency on the application—Each application program must have its own processing routines.
- Incompatible file formats (Lack of Data Sharing and Availability).
- Data Inconsistency and confusion
- Poor Security and administration

3. Database Approach

- A Database Approach is shared collection of logically related data, designed to meet the information needs of an organization. (Centralized System).
- The database approach involves the use of structured database systems (e.g., relational databases like MySQL, Oracle) to manage, store, and organize data.
- This method offers a more sophisticated, efficient, and scalable way of handling data compared to the manual or file-based approaches.
- Supports large volumes of data and allows for complex relationships between data sets.
- Provides data integrity, security, and concurrency control.
- Example: a relational database that stores customer data, order details, and inventory information, with the ability to run complex queries to generate reports.

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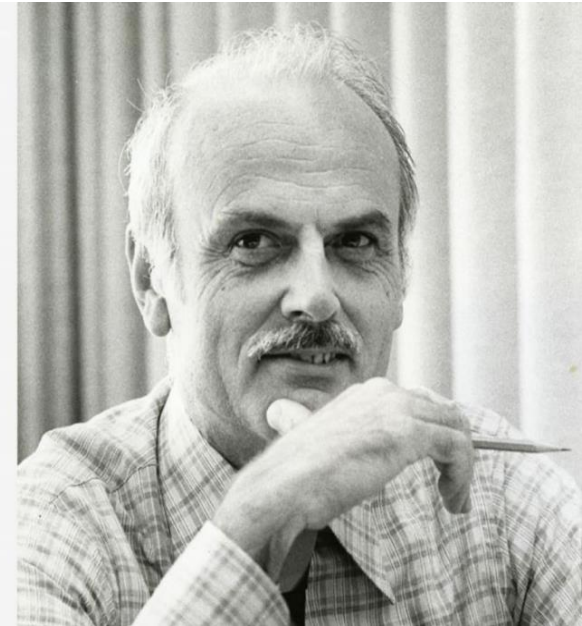
The discovery and foundation of modern database systems are often attributed to **Dr. Edgar F. Codd (1970).**----→**Relational Database.**

Key milestones in the history of the database approach:

1. Early Data Management (1960s)
2. Introduction of the Hierarchical Model (1960s)
3. Network Model (1969)
4. Relational Model (1970)
5. **First Relational Databases (1970s-1980s)**
6. Widespread Adoption (1980s-Present)

Edgar F. Codd

The Father of Relational
Database Management
Systems



Contd...

Current Trend:

- **NoSQL (Not Only SQL) databases** ---→ crucial in modern software development, especially for web-scale applications, big data analytics, and real-time systems.
- **Examples:** MongoDB, Apache Cassandra, Amazon Neptune, Hbase.

What is a Database?

- A database is a collection of related data.
- For example, the names, telephone numbers, and addresses of the people are stored in a address book.
- A database is a well organized, logically related and shared collection of data that is designed to meet the information needs of various users in an organization.
- The database is an integrated collection of facts about an *organization*. Organization can be a University, Insurance companies, Banks, Airlines, Governmental and Non-governmental Organizations etc.
- The database is used as a central data source for other applications.

Implicit properties of database:

- A database represents some aspect of the real world, sometimes called the Mini world or the universe of discourse (UoD). Changes to the mini world are reflected in the database.
- Database is a logically coherent collection of data with some inherent meaning
- A database is designed, built, and populated with data for a specific purpose. In other words, it presents the interaction with events in the real world.

What is database management system (DBMS)

- A **Database Management System (DBMS)** is **software** that provides an interface for users to interact with the database, allowing for the storage, modification, retrieval, and management of data.
- The main purpose of a DBMS is to provide **efficient, secure, and consistent management** of data.
- The main purpose of a DBMS is to provide efficient, secure, and consistent management of data.
- DBMS handles tasks like:
 - Data storage and retrieval
 - Data security and integrity
 - Backup and recovery
 - Transaction management

Example: Microsoft SQL Server, MySQL, Oracle Database, PostgreSQL, MongoDB.

What is database management system (DBMS)

An example of a large commercial database is Amazon's product catalog database. Walmart's Inventory Database, Google's Ad Database are other examples.

Characteristics of database approach

A number of characteristics distinguish the database approach from the traditional approach of programming with files.

- **Self-Describing Nature of a Database System**
- **Insulation between Programs and Data, and Data Abstraction**
- **Support of Multiple Views of the Data**
- **Sharing of Data and Multiuser Transaction Processing**

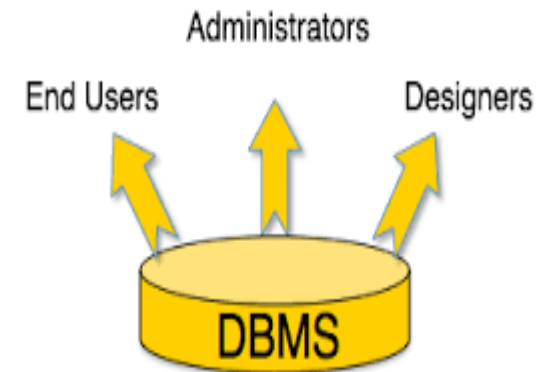
Advantages of Using the DBMS Approach

- **Controlling Redundancy**
- **Restricting Unauthorized Access**
- **Providing Persistent Storage for Program Objects**
- **Providing Storage Structures and Search Techniques for Efficient Query Processing**
- **Providing Backup and Recovery**
- **Providing Multiple User Interfaces**
- **Representing Complex Relationships among Data**
- **Enforcing Integrity Constraints**
- **Permitting Inferencing and Actions Using Rules and Triggers**

Actors on the Scene

People whose jobs involve the day-to-day **use** of a large database.

- **Database Administrators (DBA):**
 - The DBA is responsible for authorizing access to the database, for coordinating and monitoring its use, and for acquiring software and hardware resources as needed.
 - The DBA is accountable for problems such as breach of security or poor system response time.



Actors on the Scene

Database Designers:

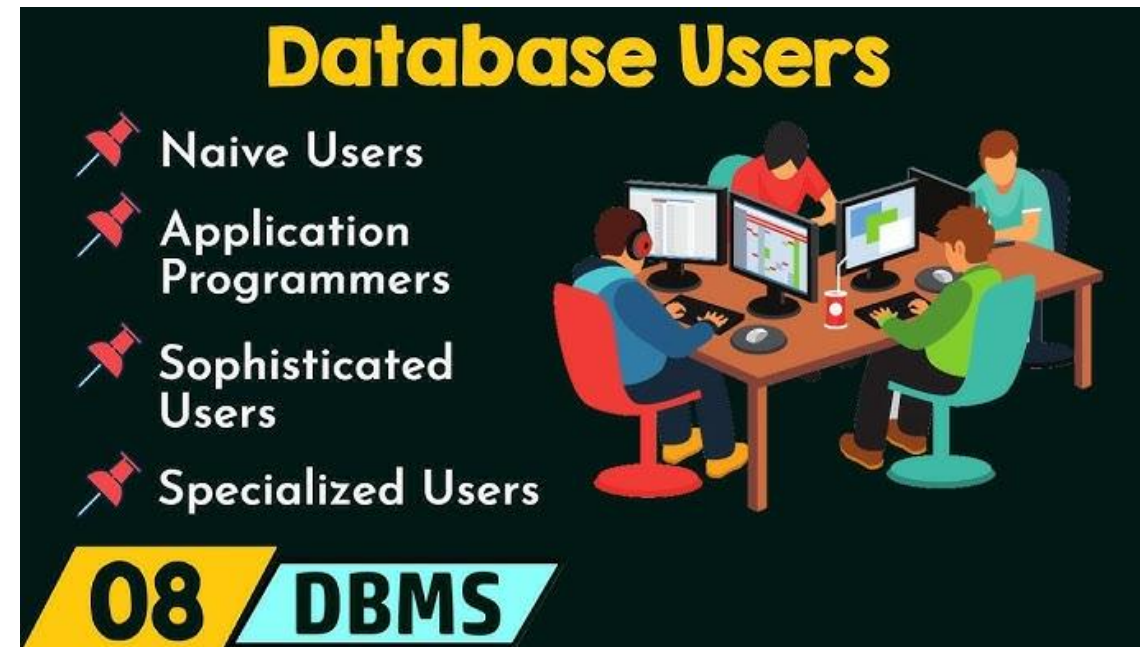
- Database designers are responsible for identifying the data to be stored in the database and for choosing appropriate structures to represent and store this data.
- These tasks are mostly undertaken before the database is actually implemented and populated with data.
- It is the responsibility of database designers to communicate with all prospective database users, in order to understand their requirements.



Actors on the Scene

End Users:

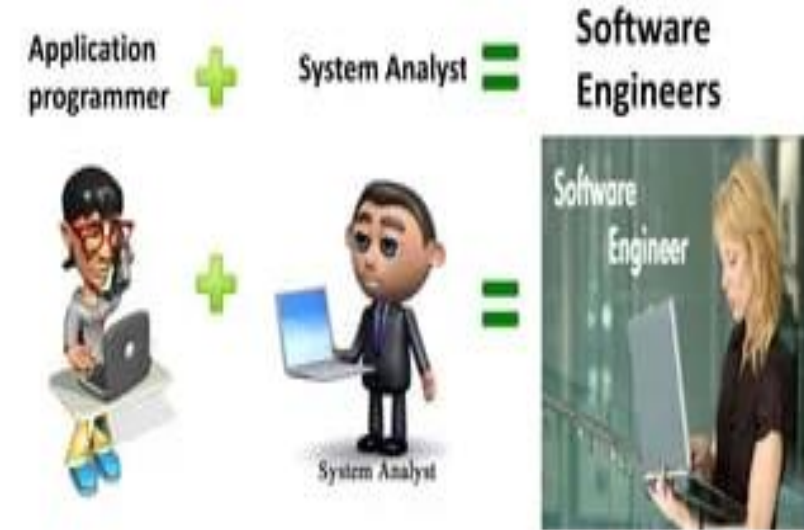
End users are the people whose jobs require access to the database for querying, updating, and generating reports.



Actors on the Scene

System Analysts and Application Programmers (Software Engineers):

- Determine the requirements of end users, especially naive and parametric end users, and develop specifications for canned transactions that meet these requirements.
- Application programmers implement these specifications as programs; then they test, debug, document, and maintain these canned transactions.
- Such analysts and programmers (nowadays called software engineers) .



Workers behind the Scene

People who are associated with the **design, development, and operation** of the DBMS software and system environment.

DBMS system designers and implementers:

- They design and implement the DBMS modules and interfaces as a software package.

Tool developers:

- They design and implement tools- are the software packages that facilitate database modeling and design, database system design, and improved performance.
- **Tools** include packages for database design, performance monitoring, natural language or graphical interfaces, prototyping, simulation, and test data generation.



Workers behind the Scene

Operators and maintenance personnel (system administration personnel):

They are responsible for the actual running and maintenance of the hardware and software environment for the database system.

Basic Database Terminologies

- Enterprise
 - an organization : A library, a bank, a university, etc.
- Entity
 - Person, place, thing, or event (property of an entity)
 - An "object" in the real world that we are interested in:
 - The object student is an entity
- Attribute (Field)
 - A character or group of characters (alphabetic or numeric), that has a specific meaning. Eg. Name, age, telephone, grade, sex, etc.

Basic Database Terminologies

- Record
 - A logically connected set of one or more Attributes that describe a person, place or thing. (Logically related data)
- File
 - A collection of related records. For example, a file might contain data about customers; or students of a certain department in a university.
- Database
 - Collection of Files|

Different database models

- **Hierarchical data model**

- Organize data in a tree structure
- There is a hierarchy of parent and child
- Data modeling to be one-to-many

- **Network data model**

- The same as hierarchical, but network model permits the modeling of many-to-many relationship
- A child can have more than one parent

Different Database Models

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- **Relational data model**
 - Stores data in the form of tables
 - A table is a collection of records
 - Each table corresponds to an entity
 - Allows the definition of data and integrity constraints
 - **Object-oriented data model**
 - Faster than relational database model
 - Applications require less code
 - Add database functionality to object programming language

When Not to Use a DBMS

In spite of the advantages of using a DBMS, there are a few situations in which a DBMS may involve unnecessary overhead costs such as:

- High initial investment in hardware, software, and training
- Overhead for providing security, concurrency control, recovery, and integrity functions
- The generality that a DBMS provides for defining and processing data (i.e. DBMS allows users to define complex data structures, relationships, and operations (e.g., querying, inserting, updating, and deleting data) in a unified, structured manner).

In particular, for **simple applications, real-time systems, small-scale projects, or resource-constrained environments**, alternatives like flat-file storage, NoSQL databases, or in-memory data stores might be more appropriate.

When Not to Use a DBMS

More desirable to use regular files for:

- Simple, well-defined database applications not expected to change at all
- Stringent, real-time requirements that may not be met because of DBMS overhead
- Embedded systems with limited storage capacity
- No multiple-user access to data

Thank You

